

Bellman Equation

Instructor

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Bellman Equation for MRPs

- Value function consists of two components:
 - Immediate reward R_{t+1}
 - Discounted cost-to-go from the next state

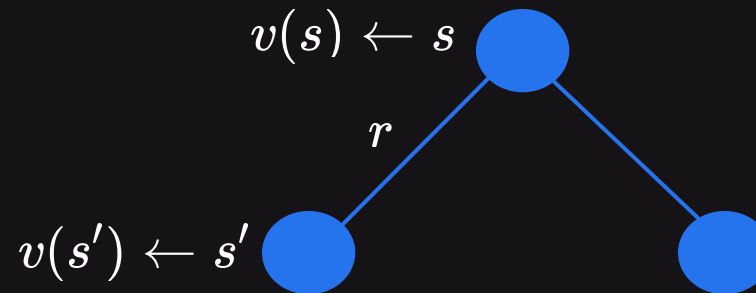
$$v(s) = \mathbb{E}[G_t \mid S_t = s]$$

$$v(s) = \mathbb{E}[R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots \mid S_t = s]$$

$$v(s) = \mathbb{E}[R_{t+1} + \gamma G_{t+1} \mid S_t = s]$$

$$v(s) = \mathbb{E}[R_{t+1} + \gamma v(S_{t+1}) \mid S_t = s]$$

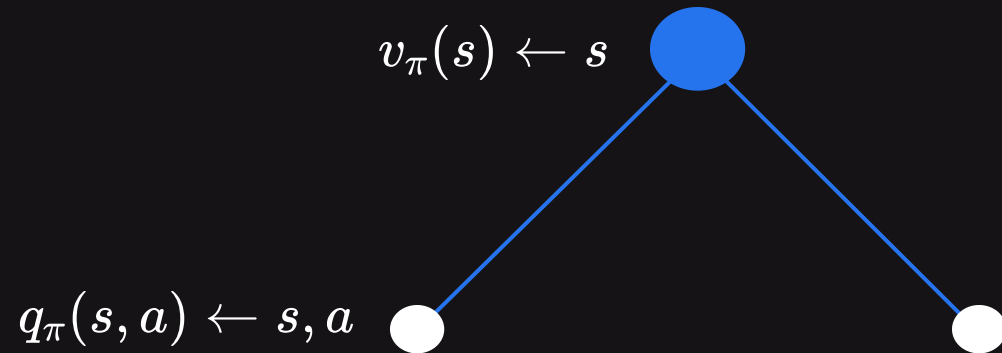
Bellman equation for MRPs



$$v(s) = R_s + \gamma \sum_{s' \in S} P_{ss'} v(s')$$

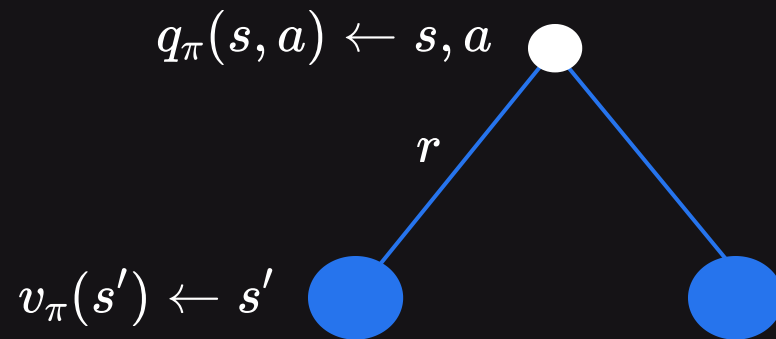
Bellman Expectation Equation

$$v_{\pi}(s) = \sum_{a \in A} \pi(a|s) q_{\pi}(s, a)$$



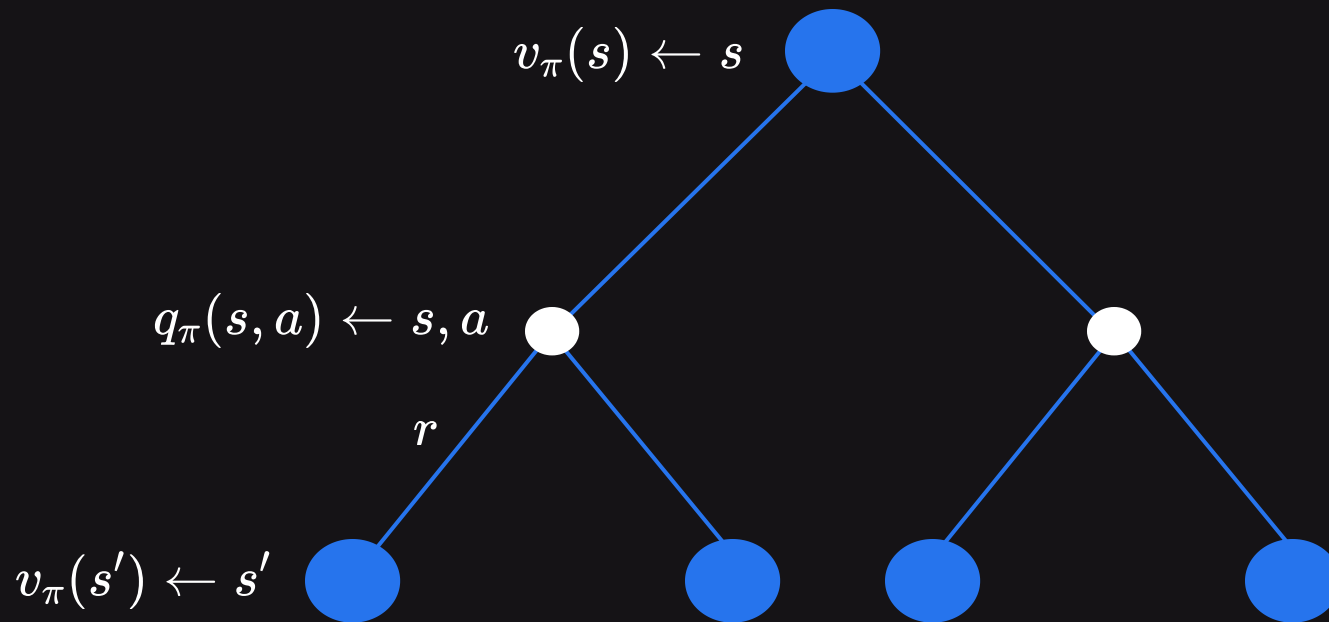
Bellman Expectation Equation

$$q_{\pi}(s, a) = R_s^a + \gamma \sum_{s' \in S} P_{ss'}^a v_{\pi}(s')$$



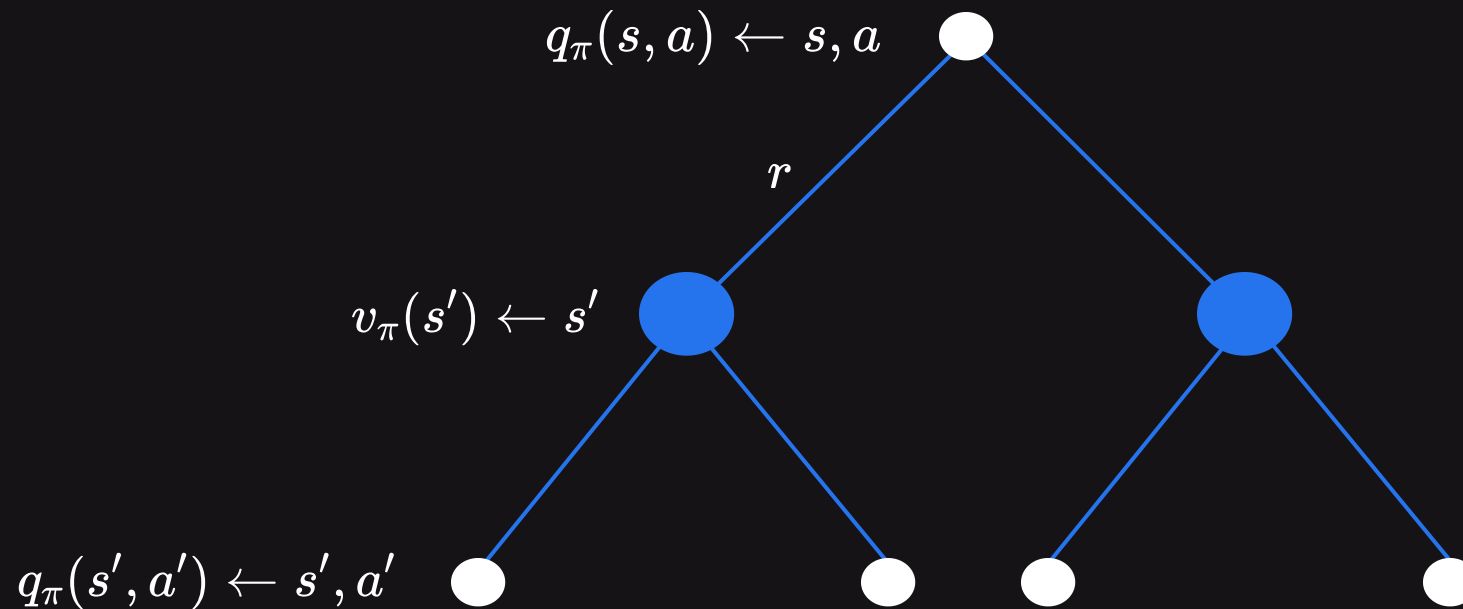
Bellman Expectation Equation for v_π

$$v_\pi(s) = \sum_{a \in A} \pi(a|s) \left(R_s^a + \gamma \sum_{s' \in S} P_{ss'}^a v_\pi(s') \right)$$



Bellman Expectation Equation for q_π

$$q_\pi(s, a) = R_s^a + \gamma \sum_{s' \in S} P_{ss'}^a \sum_{a' \in A} \pi(a'|s') q_\pi(s', a')$$



Optimal value function

An optimal policy π_* must maximize the value function

- Optimal state-value function $v_*(s)$ is the maximum value of state-value function over all policies

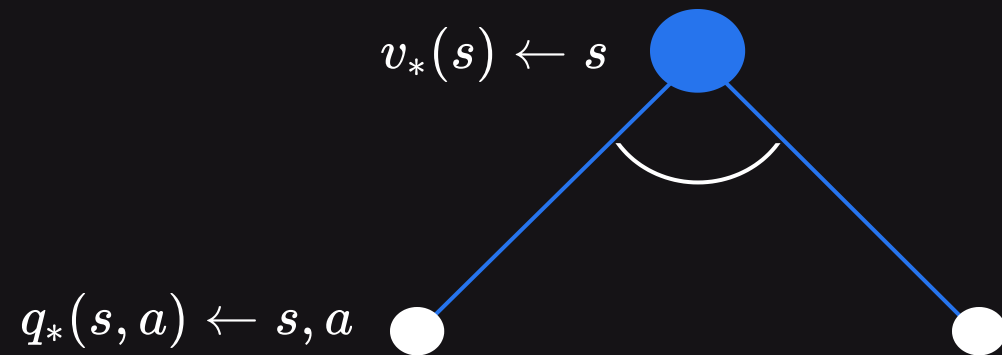
$$v_*(s) = \max_{\pi} v_{\pi}(s)$$

- Optimal action-value function $q_*(s,a)$ is the maximum value of action-value function over all policies

$$q_*(s, a) = \max_{\pi} q_{\pi}(s, a)$$

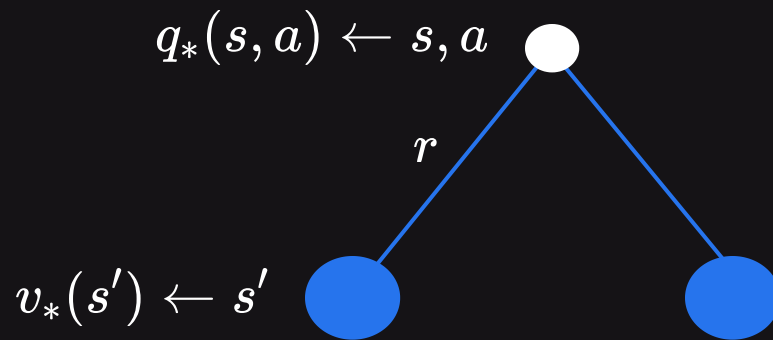
Bellman Optimality Equation

$$v_*(s) = \max_a q_*(s, a)$$



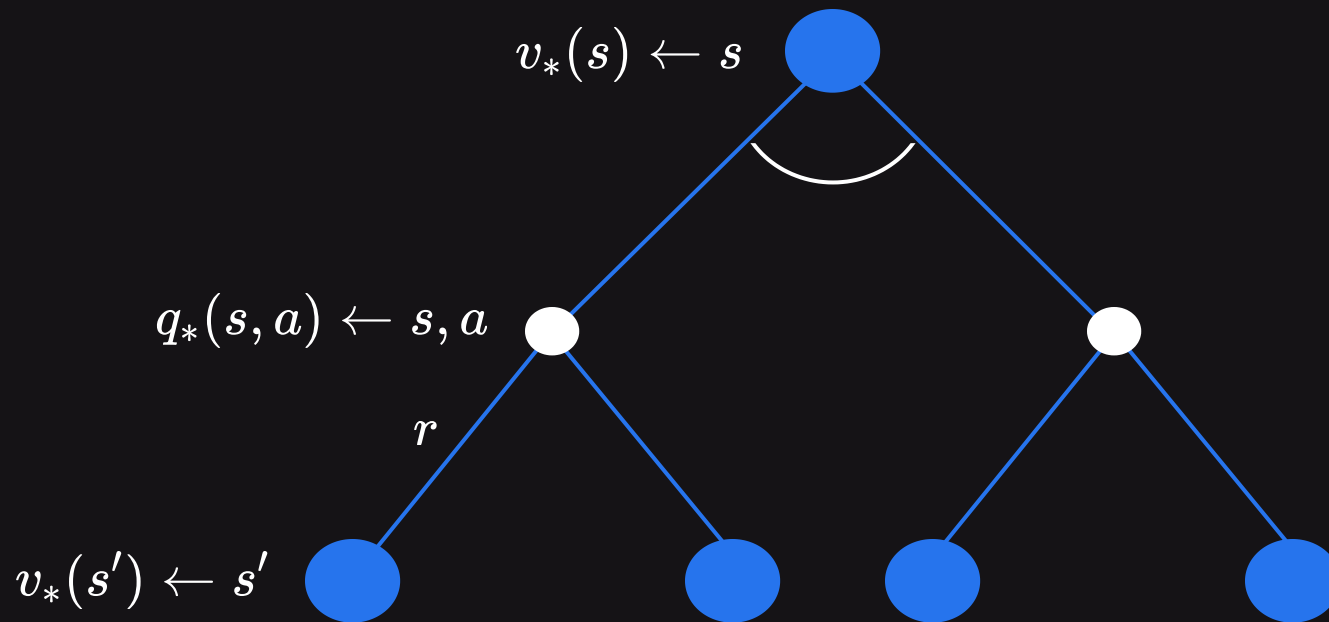
Bellman Optimality Equation

$$q_*(s, a) = R_s^a + \gamma \sum_{s' \in S} P_{ss'}^a v_*(s')$$



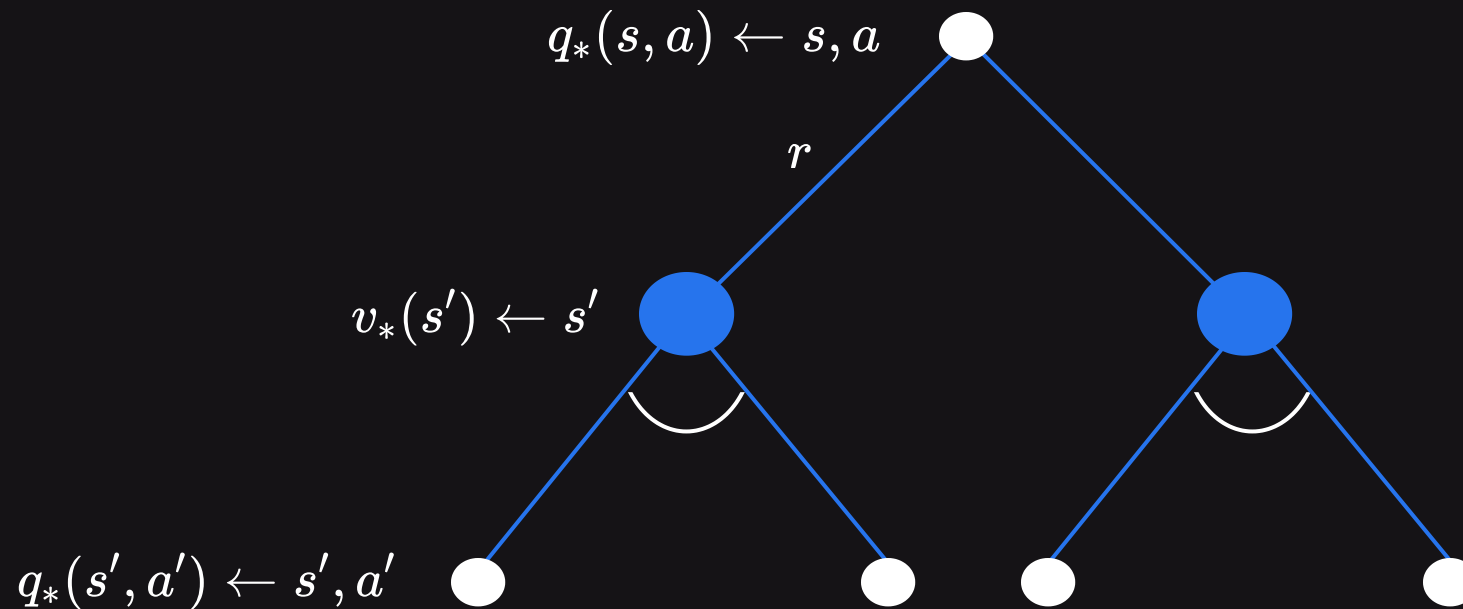
Bellman Optimality Equation for v_*

$$v_*(s) = \max_a \left(R_s^a + \gamma \sum_{s' \in S} P_{ss'}^a v_*(s') \right)$$



Bellman Optimality Equation for q_*

$$q_*(s, a) = R_s^a + \gamma \sum_{s' \in S} P_{ss'}^a \max_{a'} q_*(s', a')$$



Thank You
